Why Suppliers Might Prefer Linear and Secret Vertical Contracts?

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Abstract

We consider suppliers who endogenously choose whether the vertical contract they use with retailers, with whom they have interlocking relationships, is secret or public. If contracts are constrained to be linear, suppliers non-cooperatively choose secret contracts instead of public contracts if supplier substitutability is high or if supplier and retailer substitutability are both low. When suppliers can choose among four contract types: secret linear, public linear, secret two-part tariffs and public two-part tariffs, in any equilibrium, secret linear contracts emerge as the equilibrium if supplier and retailer substitutability are high while public two-part tariffs emerge as the equilibrium if these substitutability levels are low.

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1 Introduction

In vertical relations between suppliers and their retailers, suppliers adopt a variety of contracts. For instance, some contracts may include only a linear wholesale price.

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Others may include a two-part tariff, comprised of a wholesale price and a fixed fee. Contracts may also vary regarding whether they are public, i.e., retailers observe each other's contracts, or secret, in which each retailer cannot observe the contract offered to other retailers. While in some cases, it is reasonable to expect that exogenous constraints limit the type of contract a supplier may adopt, in other cases, suppliers may have some flexibility in choosing their contract type. This raises the question of the strategic implications of choosing one type of vertical contract over another, and how the intensity of competition among suppliers and among retailers shapes suppliers' choice of a particular type of vertical contract.

In this paper, we consider a model with two competing differentiated suppliers selling to two competing differentiated retailers. The four firms engage in "interlocking relationships", i.e., each supplier sells to both retailers and each retailer buys from both suppliers. Suppliers compete in offering vertical contracts to the two retailers and then retailers compete in prices.

We start by assuming that vertical contracts are constrained to be linear for exogenous reasons. For example, retailers may not agree to commit to paying a fixed fee if they are financially constrained, or if demand is uncertain, such that their revenues may not cover upfront fixed fees. Indeed, previous literature has demonstrated that in many industries vertical contracts are linear.[1] Given the linearity of vertical contracts, we ask whether suppliers strategically prefer to offer public or secret contracts. The model reveals that when supplier substitutability is high, both suppliers earn higher profits under secret linear contracts than under public linear contracts. When supplier substitutability is low, suppliers prefer secret contracts if retailer substitutability too is low. Moreover, under stricter conditions than the ones above with regard to supplier and retailer substitutability, suppliers strategically commit to secret linear contracts non-cooperatively. This result differs from the case of a monopolistic supplier studied by Gaudin (2019), who has shown that in such a case secret linear contracts are less

[1]Such industries include, for example, multichannel television (Crawford, and Yurukoglu (2012) and Crawford et al (2019)), contracts between hospitals and insurers (Ho and Lee (2017)), or between insurers and suppliers of medical devices (Grennan (2013, 2014)). Gilbert (2015) reported that linear pricing was common in the contracts between publishers and retailers of e-books and printed books and between Apple and music suppliers selling via iTunes (see Gaudin (2019) for these citations). Linear pricing for printed books was also documented in the case: In re Bookends & Beginnings LLC (District Court for the Southern District of New York), 2023 U.S. Dist. LEXIS 151757. Gaudin (2019) reports that linear pricing is used for access to telecom facilities. Smith and Thanassoulis (2009) report that linear contracts are used in the wholesale distribution of milk to supermarket chains. See also the industries mentioned in Inderst and Valletti (2009)'s linear pricing model.

profitable to the supplier than public linear contracts. With a secret linear contract, the supplier has an opportunistic incentive to secretly raise the wholesale price. With a monopolistic supplier, this only enhances double-marginalization and reduces the supplier's profit. By contrast, we show how with competing suppliers, this effect commits the supplier to set a high wholesale price, motivating the competing supplier to raise its own wholesale price, and hence softening upstream competition.

We also show that regardless of the level of supplier or retailer substitutability, retailers always prefer public linear contracts to secret linear contracts. This further implies that when suppliers wish to implement secret linear contracts in order to relax upstream competition, they need to ask their retailers to sign non-disclosure agreements prohibiting them from revealing their vertical contracts to each other.

We then consider the case where each supplier can strategically choose among all of the following types of contracts: 1) public two-part tariffs; 2) public linear contracts; 3) secret two-part tariffs and; 4) secret linear contracts. We find that even when all four possibilities are open to each supplier, only two equilibria emerge: When supplier and retailer substitutability are both particularly high, the unique equilibrium involves both suppliers adopting secret linear contracts. When supplier and retailer substitutability are both particularly low, the unique equilibrium involves both suppliers choosing public two-part tariffs. When supplier and retailer substitutability are intermediate, there are two possible equilibria: one with both suppliers adopting secret linear contracts and one with both of them electing public two-part tariffs.

Our result, showing circumstances when the unique equilibrium involves both suppliers committing to secret linear contracts when supplier and retailer competition is intense can explain why suppliers may strategically commit to secret linear contracts even when they could have committed to public ones, and even when two-part tariffs are feasible. The result further implies that contract secrecy, coupled with linear pricing, in and of itself, has a strategic effect, in relaxing upstream competition.

To the best of our knowledge, our paper is the first to consider secret linear contracts in a successive duopoly with interlocking relationships, suppliers' choices between secret and public vertical contracts, and suppliers' equilibrium choices when all four types of contracts are available to them.

As shown by the seminal studies of Hart and Tirole (1990), O'brien and Shaffer (1992), McAfee and Schwartz (1994) Rey and Verge (2004) and Rey and Tirole (2007), when vertical contracts are secret, suppliers face an opportunism problem, so the nature

of vertical contracts diverges from the equilibria that would have emerged under public contracts. This opportunism problem has been extended and implemented in various directions. For example, Montez (2015) shows that the opportunism problem extends to the case of make-to-stock vertical contracts and Nocke and Rey (2018) have shown it to carry over to successive duopolies with interlocking relationships and non-linear contracts. We contribute to this literature by considering the implications of suppliers' opportunism problem in a successive duopoly with secret linear contracts, and the strategic implications of contract choice being endogenous when secret linear contracts are available to suppliers.

Our paper is most related to Gaudin (2019), who studies a single supplier using linear contracts which are either secret or public. He shows that the supplier prefers public contracts to secret ones. We show that when suppliers compete with one another, their preference between public and secret contracts can reverse, because in certain industry circumstances they prefer contract secrecy as a means of softening upstream competition. Another difference between our results and those of Gaudin (2019) is that in his framework, the level of product substitutability, or intensity of competition, does not change the supplier's preference for public contracts. This is while in our framework, the level of supplier and retailer substitutability affects the question whether suppliers adopt secret or public contracts in equilibrium. Finally, unlike Gaudin (2019), who focuses on linear contracts, we ask what is the equilibrium choice of the type of contract when the four types of vertical contracts (secret linear, public linear, secret two-part tariffs and public two-part tariffs) are available to suppliers.

Our paper is also related to Pagnozzi and Picollo (2012), who study competition between two vertical hierarchies with secret two-part tariffs. They show that if retailers' beliefs are symmetric, when retailer 1 receives an offer from her supplier that is higher than expected, she believes that the competing supplier is offering a similarly high wholesale price to retailer 2, and this raises the fixed fee retailer 1 is willing to pay her supplier. Hence, they demonstrate how, despite two-part tariffs being secret, suppliers can use vertical separation between them and retailers to relax competition. Our paper is different than theirs in various ways: we study interlocking relationships while they consider two vertical hierarchies; our paper studies how the secrecy of linear contracts, under passive beliefs of retailers, relaxes upstream competition. By contrast, they consider two-part tariffs and show that, despite their secrecy, upstream competition is relaxed due to symmetric beliefs. In their model, suppliers prefer secret to public

contracts only when retailers' instruments are strategic substitutes, whereas in our model suppliers may prefer secret (linear) contracts when retailers' instruments are strategic complements.

We also contribute to the literature comparing linear contracts with two-part tariffs. Gal-Or (1991), Cachon and Kök (2010), Li et al (2013); and Feng and Lu (2013) demonstrate how whether contracts are public or secret, suppliers may prefer linear contracts over non-linear contracts because linear contracts can relax competition between suppliers when such competition is sufficiently intense, although in equilibrium, suppliers may adopt non-linear contracts even when they would be better off with linear contracts. Reisinger and Schnitzer (2012) and Pagnozzi, Piccolo and Basi (2016) study models with endogenous entry and successive duopolies. Pagnozzi, Piccolo and Basi (2016) show that in such an environment, when retailers have private information about costs, suppliers prefer linear contracts, because they induce less entry. Reisinger and Schnitzer (2012) show that absent asymmetric information, linear contracts induce more entry than two-part tariffs. Rey and Stiglitz (1995) study competing suppliers' choice between exclusive territories contracts (with linear pricing or two part tariffs, and either public or secret contracts). They show that if contracts are assumed to be public, exclusive territories with franchise fees may be a dominant strategy even though suppliers may prefer exclusive territories with linear pricing. They also show that if contracts are assumed to be secret, if two-part tariffs are ruled out, exclusive territories are better for suppliers and may be chosen in equilibrium.

We contribute to this literature by studying suppliers' choice between secret and public contracts. We also show how and when secret linear contracts or public two-part tariffs can emerge as equilibria when we make all four contract types (secret linear, public linear, secret two-part tariffs, and public two-part tariffs) available to suppliers. Cachon and Kök (2010) show that with public contracts, when two-part tariffs, linear contracts, and quantity discount contracts are available, two-part tariffs are a dominant strategy, even when linear contracts are better for suppliers. In Pagnozzi, Piccolo and Basi (2016)'s paper, when linear contracts and two-part tariffs are available, two-part tariffs are a dominant strategy, even though linear contracts are more beneficial to suppliers. We show that when secret linear contracts are available, and retailer and supplier competition is sufficiently intense, public two-part tariffs are no longer a dominant strategy, and instead secret linear contracts emerge as a unique equilibrium. The reason is that secret linear contracts are more effective than public linear contracts in relaxing

upstream competition, due to suppliers' opportunism that arises when contracts are secret. As Arya and Mittendorf (2011) show, with two-part tariffs, public contracts induce suppliers to raise wholesale prices to relax competition between their retailers, while secret contracts eliminate this incentive. Hence, with two-part tariffs, contract secrecy is pro-competitive and it reduces retail prices. By contrast, as we demonstrate, with linear vertical contracts, contract secrecy relaxes upstream competition and raises retail prices compared to the case of public contracts.

We contribute to this literature by studying suppliers' choice between secret and public contracts. We also show how, when all four contract types are available to suppliers, either secret linear contracts or public two-part tariffs can emerge in equilibrium, depending on the intensity of supplier and retailer competition.

The rest of the paper is organized as follows. Section 2 describes the model. Section 3 assumes contracts are exogenously constrained to be linear, and studies suppliers' choice between public and secret contracts. Section 4 analyzes suppliers' choices when all four types of vertical contracts (secret linear, secret two-part tariffs, public linear, and public two-part tariffs) are available. Section 6 concludes.

2 The model

We consider a market with two upstream suppliers, A and B, selling to two downstream retailers, 1 and 2. For simplicity, we normalize production and retail costs to 0.

Each supplier supplies to both retailers. The demand function faced by retailer $i = 1, 2$ for the product of supplier $j = A, B$ is: $q_{i,j}(\mathbf{p})$, where $\mathbf{p} = (p_{i,j}, p_{-i,j}, p_{i,-j}, p_{-i,-j})$ is the price vector. As our results depend on the degree of substitutability between products, and in order to obtain closed-form solutions to the degree of substitutability that determines each result, we adopt a linear demand specification. Consider a representative consumer with a utility function:

$$U = \sum_{i=1,2} \sum_{j=A,B} \left(q_{ij} - \frac{q_{ij}^2}{2} - p_{ij} q_{ij} \right)$$
$$-\sigma_S \sum_{i=1,2} q_{iA} q_{iB} - \sigma_R \sum_{j=A,B} q_{1j} q_{2j} - \sigma_S \sigma_R (q_{1A} q_{2B} + q_{2A} q_{1B}). \tag{1}$$

The parameters σ_S and σ_R measure the degree of substitutability between suppliers (products) and between retailers, respectively, where $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$.

Notice that the last term of (1) represents substitutability between product 1A (product A sold by retailer 1) and product 2B (product B sold by retailer 2) or between product 2A and product 1B. This combines two dimensions of differentiation, both between suppliers and between retailers. Hence the degree of substitutability here is lower than only between suppliers or only between retailers.

Differentiating U with respect to q_{ij}, we have that the demand function faced by retailer $i = 1, 2$ for the product of supplier $j = A, B$ is:

$$q_{ij}(\mathbf{p}) = \frac{1}{(1+\sigma_S)(1+\sigma_R)} - \frac{p_{ij} - \sigma_S p_{i,-j} - \sigma_R p_{-i,j} + \sigma_S \sigma_R p_{-i,-j}}{(1-\sigma_S^2)(1-\sigma_R^2)} \tag{2}$$

Notice that $p_{-i,-j}$ has a negative effect on $q_{ij}(\mathbf{p})$. This is driven by the fact that substitutability between product $-j$ sold by retailer $-i$ and product j sold by retailer i is significantly smaller than either the substitutability between retailers selling the same product or the substitutability between products sold by the same retailer. For example, if retailer 1 raises the price of product A, the consumer is inclined to purchase product A from retailer 2 instead, and then she will purchase less of product B from retailer 2: the consumer finds it easier to buy product A from retailer 2, and to buy product A instead of B from retailer 2, than making two switches at the same time: to retailer 2 as well as to product B.

3 Public linear contracts versus secret linear contracts

In this section, we assume vertical contracts are constrained to be linear. Given the linearity of vertical contracts, we compare public linear contracts with secret linear contracts. We show that suppliers prefer secret linear contracts to public linear contracts when supplier substitutability is high or when supplier and retailer substitutability is low. Moreover, when these characteristics (high supplier substitutability or low supplier and retailer substitutability) are even more extreme , both suppliers adopt secret linear contracts non-cooperatively. Otherwise, suppliers may find themselves in an inferior equilibrium, in which both adopt public linear contracts while they would have collectively preferred that both adopt secret linear contracts. Retailers' for their part, always prefer public linear contracts to secret ones. The next sub-section analyzes the case of public linear contracts.

3.1 Public linear contracts

Consider the case in which the suppliers' wholesale prices are public. We start with the retailers' strategies. Given the wholesale price $w_{i,j}$ supplier $j = A, B$ offers retailer $i = 1, 2$, retailer i sets $p_{i,j}$ and $p_{i,-j}$ so as to maximize:

$$\pi_i(\mathbf{p}) \equiv \sum_{j=\{A,B\}} (p_{i,j} - w_{i,j}) q_{i,j}(\mathbf{p}). \tag{3}$$

Maximizing $\pi_i(\mathbf{p})$ with respect to $p_{i,j}$ and $p_{i,-j}$, and given the symmetry between retailers, let $p_{i,j}(\mathbf{w})$ denote the equilibrium retail price given the wholesale price vector, $\mathbf{w} \equiv (w_{i,j}, w_{-i,j}, w_{i,-j}, w_{-i,-j})$, where

$$p_{i,j}(\mathbf{w}) = \frac{1 - \sigma_R}{2 - \sigma_R} + \frac{2w_{i,j} + \sigma_R w_{-i,j}}{4 - \sigma_R^2} \tag{4}$$

It is straightforward to see that retail prices increase with the wholesale prices. Moreover, given $\mathbf{w}$, retail prices are a function of the level of substitutability between retailers, σ_R. To see how, we have that when supplier j sets the same wholesale price to both retailers such that $w_{i,j} = w_{-i,j} = w_j$ (which will be the case in equilibrium), $p_{i,j}(\mathbf{w}) = \frac{1-\sigma_R+w_j}{2-\sigma_R}$. At $\sigma_R = 0$, retailers behave as monopolies and set $p_{i,j}(\mathbf{w}) = \frac{1+w_j}{2}$. As substitutability between retailers increases, the retail price decreases and converges to w_j as σ_R approaches its highest level, 1, in which case retailers are Bertrand competitors.

Supplier $j = A, B$ sets $w_{i,j}$ and $w_{-i,j}$ to maximize:

$$\pi_j(\mathbf{w}) = \sum_{i=\{1,2\}} w_{i,j} q_{i,j}(p_{ij}(\mathbf{w}), p_{-i,j}(\mathbf{w}), p_{i,-j}(\mathbf{w}), p_{-i,-j}(\mathbf{w})). \tag{5}$$

Notice that when supplier j sets its wholesale prices, it takes into account that charging retailer i a low $w_{i,j}$ decreases the rival retailer $-i$'s retail price, as retailer $-i$ expects a lower retail price charged by retailer i. This benefits the supplier, who is then able to sell a larger quantity and mitigate double-marginalization. As highlighted in the next sub-section, this incentive to set a low $w_{i,j}$ does not exist under secret contracts.

Maximizing $\pi_j(\mathbf{w})$ with respect to $w_{i,j}$ and $w_{-i,j}$, we have that the competitive wholesale prices when both suppliers adopt public linear contracts (denoted "PL") are

$w_{i,j} \equiv w^{PL,PL}$,[2] where

$$w^{PL,PL} = \frac{1-\sigma_S}{2-\sigma_S}.$$

It is straightforward to see that at $\sigma_S = 0$, $w^{PL,PL} = \frac{1}{2}$ and there is standard double marginalization involving successive monopolies. The equilibrium $w^{PL,PL}$ is decreasing with σ_S, due to the increased intensity of competition between suppliers, and approaches 0 as σ_S approaches 1, yielding Bertrand competition.

Let $\mathbf{w}^{PL,PL} \equiv (w^{PL,PL}, ..., w^{PL,PL})$. The equilibrium retail price under public linear contracts is $p^{PL,PL} \equiv p_{i,j}(\mathbf{w}^{PL,PL})$, where

$$p^{PL,PL} = 1 - \frac{1}{(2-\sigma_S)(2-\sigma_R)}.$$

The retail price $p^{PL,PL}$ is decreasing with both σ_S and σ_R. At $\sigma_S = \sigma_R = 0$, $p^{PL,PL} = \frac{3}{4}$ and $p^{PL,PL}$ approaches 0 as σ_S and σ_R approach 1.

Each supplier's equilibrium profit is:

$$\pi_S^{PL,PL} \equiv \pi_j(\mathbf{w}^{PL,PL}) = \frac{2(1-\sigma_S)}{(2-\sigma_S)^2(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)}. \tag{6}$$

Lemma 1 summarizes the effects of the substitutability parameters σ_R and σ_S on the supplier's profit:

Lemma 1. *Suppose that suppliers offer public linear contracts. A supplier's profit is decreasing with supplier substitutability, σ_S. Moreover, retailer substitutability, σ_R, decreases the supplier's profit if $\sigma_R < 1/2$ and increases the supplier's profit otherwise.*

Intuitively, as suppliers become closer substitutes, their profits decline because price competition between them intensifies. The effect of retailer substitutability on suppliers' profits, however, is non-monotonic. Intuitively, as σ_R increases, double marginalization decreases and this raises the suppliers' profits. On the other hand, when retailers become closer substitutes, variety decreases and this reduces aggregate demand. This decreases the suppliers' profits. Notice that the effect of σ_R on a supplier's profit is not affected by σ_S.

[2]Note that although we allow suppliers to discriminate between retailers, their profit-maximizing wholesale prices involve no discrimination.

Finally, let $\mathbf{p}^{PL,PL} \equiv (p^{PL,PL}, ..., p^{PL,PL})$. The retailer's profit is:

$$\pi_R^{PL,PL} = \pi_i(\mathbf{p}^{PL,PL}) = \frac{2(1-\sigma_R)}{(2-\sigma_S)^2(2-\sigma_R)^2(1+\sigma_S)(1+\sigma_R)}, \tag{7}$$

which is increasing in σ_S and decreasing in σ_R.

3.2 Secret linear contracts

This sub-section considers secret linear contracts: each retailer observes only the wholesale prices it is offered. Suppose that the equilibrium wholesale prices when both suppliers adopt secret linear contracts (denoted "SL") are $\mathbf{w}^{SL,SL} \equiv (w^{SL,SL}, w^{SL,SL}, w^{SL,SL}, w^{SL,SL})$. Consequently, the equilibrium retail prices are $\mathbf{p}^{SL,SL} \equiv (p^{SL,SL}, p^{SL,SL}, p^{SL,SL}, p^{SL,SL})$, where $p^{SL,SL} \equiv p_{i,j}(\mathbf{w}^{SL,SL}) = \frac{1-\sigma_R+w^{SL,SL}}{2-\sigma_R}$ is given by (4) after applying symmetry. We consider perfect Bayesian equilibria.

The equilibrium wholesale prices have to be immune to the following deviation: suppose that supplier j ($j = A, B$) makes a secret offer to retailer i ($i = 1, 2$) with $w_{i,j} = w^{Dev} \neq w^{SL,SL}$. Only retailer i observes this deviation, but it cannot observe the wholesale price that supplier j offered the competing retailer, $-i$. This raises the question of retailer i's belief concerning the wholesale price supplier j offers retailer $-i$. Following Hart and Tirole (1990), O'brien and Shaffer (1992) and Gaudin (2019), we assume that beliefs are *passive*: if a retailer receives an out-of-equilibrium offer, it expects that all other contract offers are not changed, i.e., that the supplier is charging the other retailer the equilibrium $w^{SL,SL}$.

Given supplier j's deviation to w^{Dev} and retailer i's beliefs that other wholesale prices are w^{Sec}, retailer i sets $p_{i,j}$ and $p_{i,-j}$ so as to maximize:

$$\pi_i(p_{i,j}, p_{i,-j}, p^{SL,SL}, p^{SL,SL}) = (p_{i,j} - w^{Dev})q_{i,j}(p_{i,j}, p_{i,-j}, p^{SL,SL}, p^{SL,SL}) \tag{8}$$

$$+(p_{i,-j} - w^{SL,SL})q_{i,-j}(p_{i,-j}, p_{i,j}, p^{SL,SL}, p^{SL,SL}).$$

Notice that retailer i expects that retailer $-i$ sets the retail price $p^{SL,SL}$ for both products A and B and expects that retailer $-i$ is paying supplier j $w^{SL,SL}$, while retailer i is paying supplier j w^{Dev}. Maximizing (8) with respect to $p_{i,j}$ and $p_{i,-j}$ yields:

$$p_{i,j}(w^{Dev}, p^{SL,SL}) = \frac{1}{2}\left(1 - \sigma_R + \sigma_R p^{SL,SL} + w^{Dev}\right), \tag{9}$$

$$p_{i,-j}(w^{SL,SL}, p^{SL,SL}) = \frac{1}{2}\left(1 - \sigma_R + \sigma_R p^{SL,SL} + w^{SL,SL}\right)$$

Supplier j sets the deviating w^{Dev} so as to maximize:

$$\begin{aligned}\pi_j(w^{Dev}, w^{SL,SL}) = w^{Dev} q_{i,j}(p_{i,j}(w^{Dev}, p^{SL,SL}), p_{i,-j}(w^{SL,SL}, p^{SL,SL}), p^{SL,SL}, p^{SL,SL}) \\ + w^{SL,SL} q_{-i,j}(p^{SL,SL}, p^{SL,SL}, p_{i,j}(w^{Dev}, p^{SL,SL}), p_{i,-j}(w^{SL,SL}, p^{SL,SL})).\end{aligned} \tag{10}$$

Maximizing $\pi_j^S(w^{Dev}, w^{SL,SL})$ with respect to the deviating w^{Dev} yields:

$$w^{Dev}(w^{SL,SL}, p^{SL,SL}) = \frac{1}{2}\left((1-\sigma_R)(1-\sigma_S) + p^{SL,SL}\sigma_R(1-\sigma_S) + (\sigma_R + \sigma_S)w^{SL,SL}\right).$$

In equilibrium, such a deviation is unprofitable for supplier j. Hence, solving $w^{Dev}(w^{SL,SL}) = w^{SL,SL}$ and $p^{SL,SL} = p_{i,j}(\mathbf{w}^{SL,SL})$ for $w^{SL,SL}$ and $p^{SL,SL}$ yields that the equilibrium wholesale and retail prices under secret linear contracts are:[3]

$$w^{SL,SL} = \frac{2(1-\sigma_S)}{4 - 2\sigma_S - \sigma_R}, \quad p^{SL,SL} = 1 - \frac{1}{4 - 2\sigma_S - \sigma_R}.$$

Comparing wholesale prices with public and secret linear contracts yields the following result:

Lemma 2. *The equilibrium wholesale price under secret linear contracts is higher than under public linear contracts: $w^{SL,SL} > w^{PL,PL}$ if $\sigma_R > 0$ and $\sigma_S < 1$. If $\sigma_R = 0$ or $\sigma_S = 1$, the equilibrium wholesale price under secret linear contracts and under public linear contracts is the same: $w^{SL,SL} = w^{PL,PL}$.*

This result is consistent with Gaudin (2019) and extends his results, which focus on a monopoly supplier, to the case of competing suppliers having interlocking vertical relations with competing retailers. The intuition for this result is similar to the monopoly case: under public linear contracts, a supplier can more effectively increase a retailer's demand for the supplier's brand by setting a low wholesale price. This is because with a low wholesale price publicly charged to both retailers, each retailer expects that the rival retailer will charge a low retail price, so both are induced to charge low retail prices. This increases retailers' demand for the supplier's brand and reduces double-marginalization. This effect does not exist with secret contracts and passive

[3] We have also verified that these prices prevent supplier j from deviating in its contract offers to both retailers simultaneously (i.e., deviating simultaneously with regard to both $w_{i,j}$ and $w_{-i,j}$).

beliefs. Here, a supplier opportunistically wishes to charge higher wholesale prices, because each retailer is reluctant to pass on the price increase, believing the rival retailer will not raise its retail price.

Notice that opportunism occurs only if $\sigma_R > 0$. If $\sigma_R = 0$, equilibrium wholesale prices are the same under secret linear contracts and under public linear contracts. Intuitively, if there is no retailer substitutability, retailers do not compete and therefore a retailer is not reluctant to pass on an increase in the wholesale price, so suppliers cannot behave opportunistically. Moreover, opportunism occurs only if $\sigma_S < 1$, because at $\sigma_S = 1$, suppliers are homogeneous Bertrand competitors and therefore do not have the market power to behave opportunistically.

With a monopoly supplier, as shown by Gaudin (2019), this effect of secret contracts with linear pricing hurts the supplier. It earns lower profits under secret linear contracts than under public linear contracts. In contrast, we find that competing suppliers may earn higher profits when both use secret linear contracts instead of public ones: each supplier benefits when the competing supplier raises its wholesale price, so suppliers' above-mentioned opportunism profitably softens price competition between suppliers. The same reasoning implies that retailers earn higher profits with public linear contracts than with secret ones.

Under secret linear contracts, each supplier's profit is:

$$\pi_S^{SL,SL} \equiv \pi_j(w^{SL,SL}, w^{SL,SL}) = \frac{4(1-\sigma_S)}{(1+\sigma_S)(1+\sigma_R)(4-2\sigma_S+\sigma_R)^2}. \tag{11}$$

Let $\mathbf{p}^{SL,SL} \equiv (p^{SL,SL}, ..., p^{SL,SL})$. The retailer's profit is:

$$\pi_R^{SL,SL} = \pi_i(\mathbf{p}^{SL,SL}) = \frac{2(1-\sigma_R)}{(1+\sigma_S)(1+\sigma_R)(4-2\sigma_S-\sigma_R)}.$$

It is straightforward to show that retailers earn lower profits under secret linear contracts than under public linear contracts. The following proposition compares the suppliers' profits under secret and public linear contracts:

Proposition 1. $\pi_S^{SL,SL} > \pi_S^{PL,PL}$ *if and only if* $\sigma_S > \overline{\sigma}_S \equiv 1 - \sqrt{1 - \frac{\sigma_R}{2}}$, *i.e.:*

i) If supplier substitutability is high ($\sigma_S \geq 1 - \frac{1}{\sqrt{2}}$), *suppliers earn higher profits under secret linear contracts than under public linear contracts regardless of retailer substitutability.*

ii) If supplier substitutability is low ($\sigma_S < 1 - \frac{1}{\sqrt{2}}$), *suppliers earn higher profits under*

secret linear contracts than under public linear contracts if retailer substitutability is low ($\sigma_R < 2\sigma_S(2 - \sigma_S)$).

iii) Retailers earn lower profits under secret linear contracts than under public linear contracts.

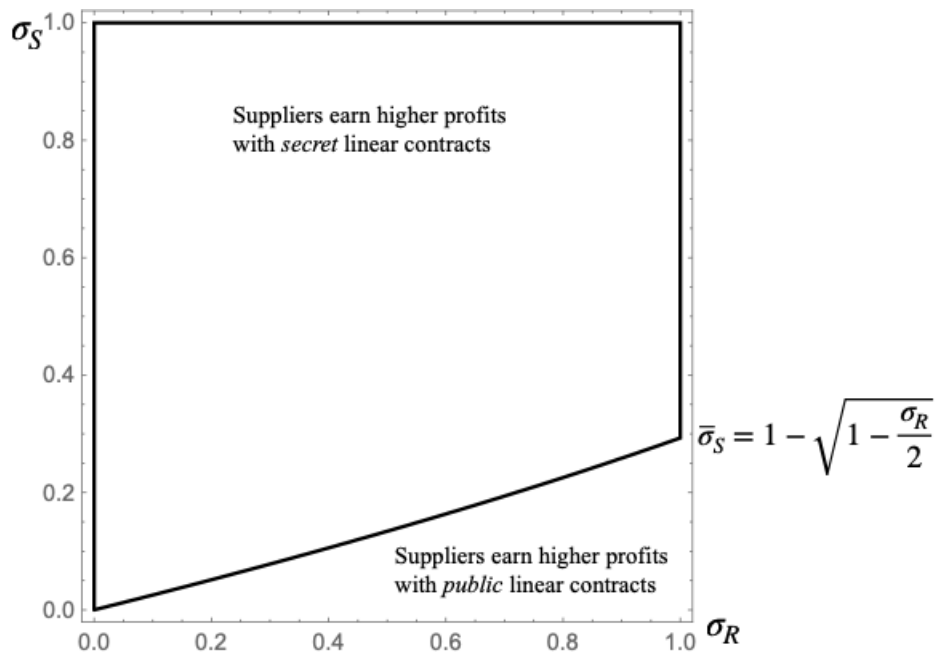

Figure 1: The regions in which suppliers prefer secret linear contracts

Figure 1 illustrates $\overline{\sigma}_S$, the threshold above which supplier substitutability needs to be for suppliers to prefer secret linear contracts, as a function of σ_R. The figure shows that if supplier substitutability exceeds a certain threshold (where $\overline{\sigma}_S$ reaches its peak, for $\sigma_R = 1$), suppliers prefer secret to public contracts for any level of retailer substitutability. However, when supplier substitutability is lower than this threshold, retailer substitutability needs to be sufficiently low for suppliers to prefer secret contracts.

Intuitively, the only benefit of secret contracts for suppliers is relaxing upstream competition. Otherwise, we know from Gaudin (2019) that a monopoly supplier prefers public contracts, because secret contracts and opportunism exacerbate double marginalization. So, when supplier substitutability is sufficiently high, suppliers always wish to relax upstream competition by adopting secret contracts. Conversely, when supplier substitutability is low, there is less upstream competition to relax, and double marginalization becomes more of a concern to suppliers. This is why in this region, retailer substitutability needs to be small for suppliers to prefer secret to public contracts. Secret contracts cause opportunism and therefore exacerbate double marginalization.

But for sufficiently small retailer substitutability, opportunism is relatively small (see part (i) of Lemma 3 below), so suppliers find it worthwhile to relax upstream competition by using secret contracts.

The third part of Proposition 1, according to which retailers always prefer public linear contracts to secret linear contracts, implies that in order to implement contract secrecy, suppliers need to use non-disclosure agreements in their relations with retailers. Otherwise, retailers would have an incentive to credibly make their contracts with suppliers observable. This further implies that non-disclosure agreements should be more prevalent when supplier substitutability is high or when supplier and retailer substitutability are low.

We can gain more insight on the driving forces of Proposition 1 by highlighting the following result: there is a strong link between suppliers' preference for secret linear contracts and the magnitude of opportunism. The level of opportunism brought about by secret contracts, denoted below as $OP(\sigma_R, \sigma_S)$, can be expressed as the increase in the wholesale price caused by secret contracts: $OP(\sigma_R, \sigma_S) \equiv w^{SL,SL} - w^{PL,PL}$. The following lemma characterizes a few important features of $OP(\sigma_R, \sigma_S)$:

Lemma 3. *(i) The level of opportunism caused by secret linear contracts, $OP(\sigma_R, \sigma_S)$, is increasing in retailer substitutability σ_R.*

(ii) For any σ_R, the degree of supplier substitutability σ_S maximizing opportunism is uniquely given by $\overline{\sigma}_S$ (the threshold from Proposition 1 above which suppliers prefer secret linear contracts).

(iii) The unique point maximizing opportunism in the (σ_R, σ_S) space is $(1, \overline{\sigma}_S(1))$.

The first part of Lemma 3 follows because the higher is retailer substitutability, the more reluctant is a retailer to pass on an increase in the wholesale price, and this exacerbates supplier opportunism. The second part of Lemma 3 implies that the line dividing the region in figure 1 where suppliers prefer secret contracts and the region where they prefer public ones is exactly the line in which, given retailer substitutability supplier substitutability (σ_S) maximizes opportunism. Along this line, suppliers are indifferent between secret and public contracts. This implies that although the attractiveness of secret contracts is driven by opportunism, suppliers need opportunism to be sufficiently moderate for them to strictly prefer secret to public contracts. This is because opportunism also imposes costs due to double marginalization. Accordingly, suppliers also need upstream competition to be sufficiently intense so that relaxing

such competition is worthwhile. For any given level of retailer substitutability σ_R, increasing σ_S above the line $\overline{\sigma}_S$ (see figure 1) makes secret contracts more attractive by reducing opportunism below its maximum level (by the uniqueness of the maximum point $\overline{\sigma}_S(\sigma_R)$) and hence reducing double marginalization. Moreover, the increase in σ_S makes upstream competition more intense, thereby increasing the need to relax such competition. Conversely, reducing σ_S below the line $\overline{\sigma}_S$ also reduces opportunism (and hence reduces double marginalization), but it also decreases the need to relax upstream competition, because such competition is softer to begin with.

The third part of Lemma 3 follows from the first two parts: the line $\overline{\sigma}_S$ maximizes opportunism for any given σ_R, while opportunism is strictly increasing in σ_R, so the point globally maximizing opportunism is on the line $\overline{\sigma}_S(\sigma_R)$ when $\sigma_R = 1$.

Under the conditions in parts (i) and (ii) of Proposition 1, had suppliers been able to coordinate with each other whether to offer secret or public linear contracts, they would have agreed to offer secret contracts, and this would harm competition between them. Yet, it is reasonable to assume that suppliers would not be allowed to coordinate in this fashion, for antitrust reasons. This raises the question of whether secret contracts are expected to arise in a non-cooperative equilibrium between suppliers. We pursue this in the next sub-section.

3.3 Choice between public and secret linear contracts in a non-cooperative game

Suppose that each supplier can non-cooperatively commit to offering public contracts. Accordingly, in the first stage, each supplier declares whether it adopts a policy of public or secret linear contracts and this policy becomes common knowledge (naturally, if a supplier chooses not to commit to public contracts, it becomes common knowledge that it chose secret contracts). As mentioned above, a commitment to secrecy can be achieved, for its part, via non-disclosure agreements between a supplier and retailers. In the second stage, suppliers set their wholesale prices. Finally, retailers compete in retail prices.

To solve the game, we identify the second and third stage equilibrium when in the first stage, only one of the suppliers, supplier j, adopts secret linear contracts, while the competing supplier $-j$ adopts public linear contracts. Define the equilibrium wholesale prices in this scenario as $w_j^{SL,PL}$ for the secret wholesale price of supplier j, and $w_{-j}^{SL,PL}$ for the public wholesale price of supplier $-j$. Supplier $-j$ expects that both retail-

ers set the retail prices $p_{ij}(\mathbf{w}^{SL,PL})$, where $\mathbf{w}^{SL,PL} = (w_j^{SL,PL}, w_j^{SL,PL}, w_{-j}^{SL,PL}, w_{-j}^{SL,PL})$. Substituting $\mathbf{w}^{SL,PL} = (w_j^{SL,PL}, w_j^{SL,PL}, w_{-j}^{SL,PL}, w_{-j}^{SL,PL})$ into the retailers' prices in (4) yields:

$$p_{i,j}(\mathbf{w}^{SL,PL}) = \frac{1-\sigma_R}{2-\sigma_R} + \frac{2w_j^{SL,PL} + \sigma_R w_j^{SL,PL}}{4-\sigma_R^2}. \tag{12}$$

Supplier $-j$ sets $w_{-j}^{SL,PL}$ so as to maximize $\pi_{-j}(\mathbf{w}^{SL,PL})$, as given by (5).

Turning to supplier j, who makes secret offers, suppose that this supplier deviates from $w_j^{SL,PL}$ by offering retailer i some $\widetilde{w}_{i,j}^{Dev} \neq w_j^{SL,PL}$. Retailer i, who expects that retailer $-i$ sets $p_{-i,j}(\mathbf{w}^{SL,PL})$ as given by (12), sets the retail prices for products j and $-j$:

$$p_{i,j}(\widetilde{w}_{i,j}^{Dev}, w_j^{SL,PL}, w_{-j}^{SL,PL}) = \frac{4(1+\widetilde{w}_{i,j}^{Dev}) - \sigma_R(2+\sigma_R(2+\widetilde{w}_{i,j}^{Dev} - w_j^{SL,PL}) - 2w_{-j}^{SL,PL})}{2(4-\sigma_R^2)},$$

$$p_{i,-j}(\widetilde{w}_{i,j}^{Dev}, w_j^{SL,PL}, w_{-j}^{SL,PL}) = \frac{\sigma_R(w_j^{SL,PL} - \sigma_R - 1) + 2(1+w_j^{SL,PL})}{4-\sigma_R^2}.$$

Supplier j sets the deviating $\widetilde{w}_{i,j}^{Dev}$ so as to maximize

$$\pi_j^S(\widetilde{w}_{i,j}^{Dev}, w_j^{SL,PL}, w_{-j}^{SL,PL}) = \widetilde{w}_{i,j}^{Dev} q_{i,j}(\cdot) + w_j^{SL,PL} q_{-i,j}(\cdot), \tag{13}$$

where the retail prices are $p_{i,j}(\widetilde{w}_{i,j}^{Dev}, w_j^{SL,PL}, w_{-j}^{SL,PL})$, $p_{i,-j}(\widetilde{w}_{i,j}^{Dev}, w_j^{SL,PL}, w_{-j}^{SL,PL})$, $p_{-i,j}(\mathbf{w}^{SL,PL})$ and $p_{-i,-j}(\mathbf{w}^{SL,PL})$. This equilibrium is immune to such a deviation when the $\widetilde{w}_{i,j}^{Dev}$ that maximizes (13) satisfies $\widetilde{w}_{i,j}^{Dev} = w_j^{SL,PL}$.

Simultaneously maximizing $\pi_{-j}(\mathbf{w}^{SL,PL})$ (as given by (5)) with respect to $w_{-j}^{SL,PL}$, and (13) with respect to $\widetilde{w}_{i,j}^{Dev}$ while imposing $\widetilde{w}_{i,j}^{Dev} = w_j^{SL,PL}$ yields that the equilibrium wholesale prices are:

$$w_j^{SL,PL} = \frac{(1-\sigma_S)(2+\sigma_S)}{4-\sigma_S^2-\sigma_R}, \quad w_{-j}^{SL,PL} = \frac{(1-\sigma_S)(4+2\sigma_S-\sigma_R)}{2(4-\sigma_S^2-\sigma_R)}.$$

The following lemma compares these wholesale prices with the wholesale prices when both suppliers adopt either public linear contracts ($w^{PL,PL}$) or secret linear contracts ($w^{SL,SL}$):

Lemma 4. *Suppose only one of the suppliers, supplier j, adopts secret linear contracts, while the competing supplier $-j$ adopts public linear contracts. For all $\sigma_R, \sigma_S \in (0,1)$, $w^{SL,SL} > w_j^{SL,PL} > w_{-j}^{SL,PL}$.*

That is, the supplier who offers secret linear contracts charges a higher wholesale price than the supplier who offers public linear contracts.[4] Moreover, both of these wholesale prices are lower than the wholesale prices when both suppliers offer secret linear contracts, and higher than the wholesale prices when both suppliers offer public linear contracts.

Substituting $w_j^{SL,PL}$ and $w_{-j}^{SL,PL}$ into the suppliers' profits in (5) yields that the equilibrium profits when supplier j offers secret linear contracts and supplier $-j$ offers public linear contracts, respectively, are:

$$\pi_j^{SL,PL} = \frac{(1-\sigma_S)(2+\sigma_S^2)}{(4-\sigma_S^2-\sigma_R)^2(1+\sigma_S)(1+\sigma_R)}, \tag{14}$$

$$\pi_{-j}^{SL,PL} = \frac{(1-\sigma_S)(4-2\sigma_S+\sigma_R)^2}{2(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)(4-\sigma_S^2-\sigma_R)^2}.$$

It is possible to show that $\pi_{-j}^{SL,PL} > \pi_j^{SL,PL}$: supplier $-j$, who offers public linear contracts, earns higher profits than supplier j, who offers secret linear contracts. Supplier $-j$, who offers public linear contracts, benefits from eliminating the opportunism problem associated with setting high wholesale prices. In addition, this supplier benefits from the high wholesale price that supplier j (who offered secret linear contracts) sets, yielding supplier $-j$ a competitive advantage.

Hence each supplier j would strategically prefer that the rival supplier, $-j$, offer secret contracts. As we show below, whether this strategic effect prevents suppliers from reaching the most profitable outcome depends on the levels of substitutability:

Lemma 5. *Suppose that supplier $-j$ offered public linear contracts. $\pi_j^{SL,PL} > \pi^{PL,PL}$ (i.e., supplier j prefers secret linear contracts) if and only if $\sigma_S > \sqrt{2-\sqrt{4-2\sigma_R}}$*

According to Lemma 5, when supplier $-j$ offers public linear contracts, it is profitable for supplier j to offer secret linear contracts if and only if supplier substitutability exceeds a threshold that is increasing with retailer substitutability. If supplier j joins supplier $-j$ in offering public linear contracts, this intensifies upstream competition and erodes profits, but then supplier j has no competitive disadvantage. If, on the other hand, supplier j offers secret linear contracts while supplier $-j$ offers public ones, competition is softened, but supplier j suffers from a competitive disadvantage, as its wholesale price will be higher than that of supplier $-j$. Notice that the threshold in

[4] We have verified that supplier j sells a positive quantity even though it charges a higher wholesale price.

Lemma 5 shares the characteristic of the threshold in Proposition 1, in that if supplier substitutability, σ_S, is very high (in particular, $\sigma_S > \sqrt{2-\sqrt{2}}$, substituting $\sigma_R = 1$ in the threshold, see the upper region of figure 2) then supplier j is always ready to offer secret contracts even when supplier $-j$ offers public ones. For lower levels of supplier substitutability, retailer substitutability too needs to be low for this to occur. With low retailer substitutability, the competitive disadvantage supplier j experiences is modest, because supplier j's opportunism problem is less severe with a low σ_R: low retailer substitutability implies that retailers are less reluctant to pass on supplier j's higher wholesale price.

Consider next the case in which supplier $-j$ offers secret linear contracts:

Lemma 6. *Suppose supplier $-j$ offers secret linear contracts. Supplier j prefers to offer secret contracts as well instead of public linear contracts if and only if $\sigma_S > \frac{1}{2}\sqrt{2-\sqrt{4-2\sigma_R}(4-\sigma_R)}$.*

Lemma 6 shows that when supplier $-j$ chooses secret linear contracts, supplier j chooses secret linear contracts too if and only if supplier substitutability is higher than a threshold that is increasing in retailer substitutability, but at a lower pace than the threshold from Lemma 5 (see figure 2). For supplier j, choosing public linear contracts exploits supplier $-j$'s vulnerability and gains supplier j a competitive advantage, but with more intense competition. On the other hand, joining supplier $-j$ and choosing secret linear contracts entails no competitive advantage, but further softens upstream competition. Here too, for sufficiently high supplier substitutability it is always worthwhile for supplier j to join supplier $-j$ and adopt secret linear contracts. But for lower supplier substitutability, retailer substitutability too needs to be sufficiently low for this to occur. With low retailer substitutability, the competitive advantage of not softening competition isn't too great (due to opportunism not being too great), so supplier j prefers to join supplier $-j$ in adopting secret linear contracts.

Combining Lemmas 3 and 4 yields the following result:

Proposition 2. *Suppose that suppliers can commit to public or secret linear contracts. Then:*

(i) *if $\sqrt{2-\sqrt{4-2\sigma_R}} < \sigma_S < 1$, there is a unique equilibrium in which both suppliers choose secret linear contracts,*

(ii) *if* $\frac{1}{2}\sqrt{2-\sqrt{4-2\sigma_R}(4-\sigma_R)} \leq \sigma_S \leq \sqrt{2-\sqrt{4-2\sigma_R}}$, *there are two equilibria, one with both suppliers choosing secret linear contracts and one with both choosing public linear contracts,*

(iii) *if* $0 < \sigma_S < \frac{1}{2}\sqrt{2-\sqrt{4-2\sigma_R}(4-\sigma_R)}$, *there is a unique equilibrium with both suppliers choosing public linear contracts.*

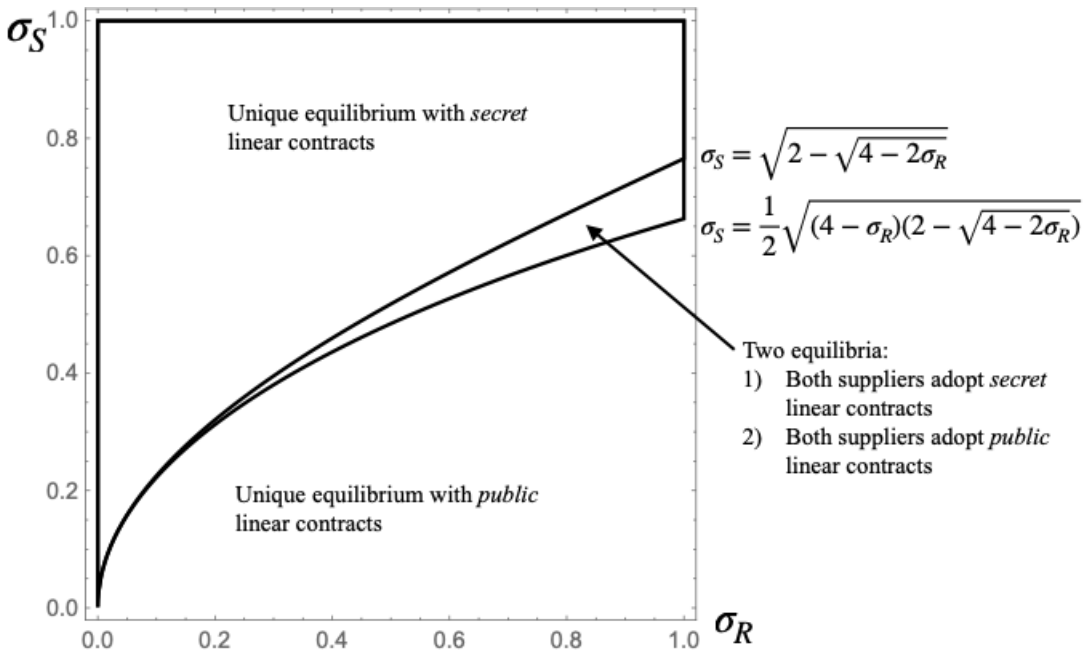

Figure 2: The equilibria between public or secret linear contracts

Figure 2 illustrates the results of Proposition 2. The upper region corresponds to the case where there is a unique equilibrium with secret linear contracts. This corresponds either to very high supplier substitutability ($\sigma_S \geq \sqrt{2-\sqrt{2}}$), implying that it is very important to relax upstream competition, or somewhat lower supplier substitutability, coupled with sufficiently low retailer substitutability. The middle region corresponds to the case where the equilibrium can involve either secret or public contracts. In the bottom region, where supplier substitutability is low, but retailer substitutability is not too low, there is a unique equilibrium with public contracts. Here each supplier defects to offering public linear contracts, to avoid the competitive disadvantage stemming from increased double marginalization in the sale of its own brand.

Proposition 2 can explain why suppliers may want to offer secret linear contracts (or, equivalently, why suppliers may not commit to using public linear contracts). Yet, in the ranges corresponding to parts 2 and 3 of Proposition 2, the non-cooperative nature

of the game may cause the resulting equilibrium to depart from what is optimal for the suppliers. Proposition 1 shows that suppliers' profits are higher when both of them offer secret contracts for any $\sigma_S > 1 - \sqrt{1 - \frac{\sigma_R}{2}}$, which is a considerably smaller threshold. Figure 2 illustrates this region, in which both suppliers prefer secret linear contracts to public linear ones. In the region $1 - \sqrt{1 - \frac{\sigma_R}{2}} < \sigma_S < \frac{1}{2}\sqrt{2 - \sqrt{4 - 2\sigma_R}(4 - \sigma_R)}$, while both suppliers are better off with secret linear contracts, the unique equilibrium, by part 3 of Proposition 2, involves both of them adopting public linear contracts. In this region, each supplier's incentive to be the only one offering public linear contracts and thereby gain a competitive advantage creates a prisoner's dilemma: both suppliers ultimately offer public linear contracts, resulting in lower profits for both. Consumers and retailers, however, gain from this prisoner's dilemma, because the equilibrium with public linear contracts involves lower wholesale and retail prices.

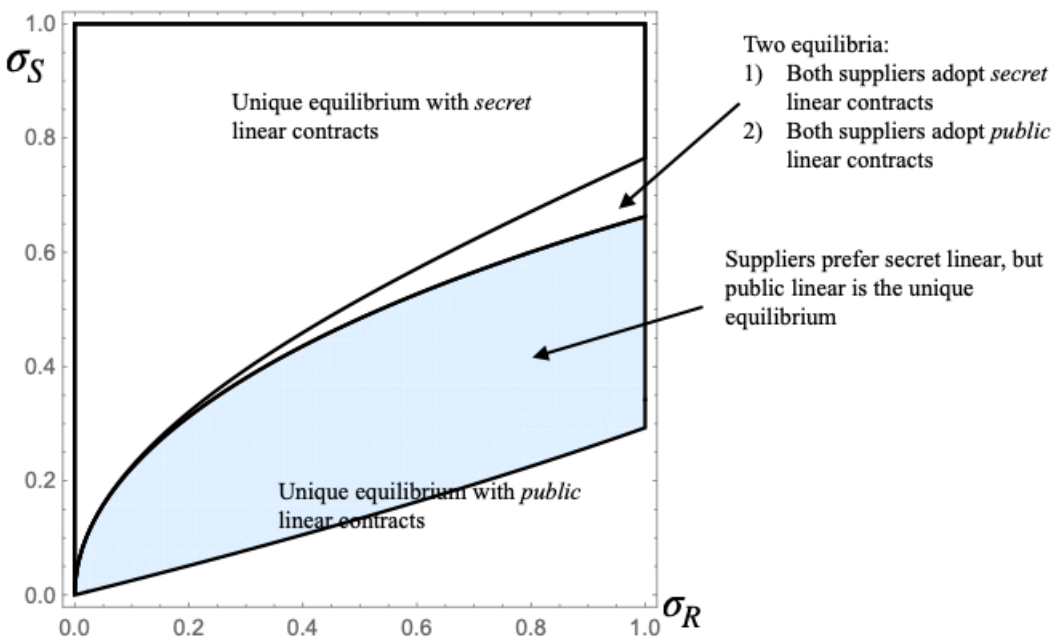

Figure 3: Prisoner's dilemma

4 Suppliers' choice of the type of contract when all options are available

Section 3 assumed contracts are linear for exogenous reasons and studied suppliers' choice between public linear contracts and secret linear contracts. As mentioned in the introduction, previous literature has studied suppliers' choice between linear contracts

and non-linear contracts when contracts are exogenously assumed to be either public or secret. In the current section, we allow suppliers to choose between all four types of contracts: secret linear; secret two-part tariffs; public linear and public two-part tariffs. Table 1 includes the suppliers' profits under each contract combination.

Supplier	Contract	Profit
j	public linear	$\frac{2(1-\sigma_S)}{(2-\sigma_S)^2(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)}$
$-j$	public linear	$\frac{2(1-\sigma_S)}{(2-\sigma_S)^2(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)}$
j	secret linear	$\frac{4(1-\sigma_S)}{(1+\sigma_S)(1+\sigma_R)(4-2\sigma_S+\sigma_R)^2}$
$-j$	secret linear	$\frac{4(1-\sigma_S)}{(1+\sigma_S)(1+\sigma_R)(4-2\sigma_S+\sigma_R)^2}$
j	secret two-part tariff	$2\frac{(1-\sigma_S)(1-\sigma_R)}{(1+\sigma_S)(2-\sigma_R)^2(1+\sigma_R)}$
$-j$	secret two-part tariff	$2\frac{(1-\sigma_S)(1-\sigma_R)}{(1+\sigma_S)(2-\sigma_R)^2(1+\sigma_R)}$
j	secret linear	$\frac{(1-\sigma_S)(2+\sigma_S)^2}{(1+\sigma_S)(1+\sigma_R)(4-\sigma_S^2+\sigma_R)^2}$
$-j$	public linear	$\frac{(1-\sigma_S)(4+2\sigma_S-\sigma_R)^2}{2(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)(4-\sigma_S^2-\sigma_R)^2}$
j	secret two-part tariff	$\frac{2(1-\sigma_S)(1-\sigma_R)(4+2\sigma_S-\sigma_R)^2}{(1+\sigma_S)(1+\sigma_R)(8-6\sigma_R+\sigma_R^2)^2}$
$-j$	secret linear	$\frac{4(1-\sigma_S)}{(1+\sigma_S)(4-\sigma_R)^2(1+\sigma_R)}$
j	public two-part tariff	$\frac{2(1-\sigma_S)}{(1+\sigma_S)(1+\sigma_S)(2-\sigma_S\sigma_R)^2}$
$-j$	public two-part tariff	$\frac{2(1-\sigma_S)}{(1+\sigma_S)(1+\sigma_S)(2-\sigma_S\sigma_R)^2}$
j	public two-part tariff	$\frac{2(1-\sigma_S)(2+\sigma_S)^2}{(1+\sigma_S)(1+\sigma_R)(4-\sigma_S^2\sigma_R)^2}$
$-j$	public linear	$\frac{2(1-\sigma_S)(2+\sigma_S\sigma_R)^2}{(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)(4-\sigma_S^2\sigma_R)^2}$
j	public two-part tariff	$\frac{(1-\sigma_S)(4+2\sigma_S+\sigma_R)^2}{2(1+\sigma_S)(1+\sigma_R)(4-\sigma_R+\sigma_S^2\sigma_R)^2}$
$-j$	secret linear	$\frac{(1-\sigma_S)(2+\sigma_S\sigma_R)^2}{(1+\sigma_S)(1+\sigma_R)(4-\sigma_R+\sigma_S^2\sigma_R)^2}$
j	public two-part tariff	$\frac{(1-\sigma_S)}{2(1+\sigma_S)(1+\sigma_R)}$
$-j$	secret two-part tariff	$\frac{2(1-\sigma_S)(1-\sigma)_R(2+\sigma_S\sigma_R)^2}{2(1+\sigma_S)(2-\sigma_R)^2(1+\sigma_R)}$
j	secret two-part tariff	$\frac{(1-\sigma_S)(2+\sigma_S)^2(1-\sigma_R)}{2(1+\sigma_S)(2-\sigma_R)^2(1+\sigma_R)}$
$-j$	public linear	$\frac{(1-\sigma_S)}{2(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)}$

Table 1: The suppliers' profits under all four contract combinations

We examine which combination emerges in a non-cooperative equilibrium. We find that in such an equilibrium, both suppliers elect either secret linear contracts or public two part tariffs, depending on the degree of supplier and retailer substitutability.

As depicted in Figure 4, in the right upper region , where supplier and retailer substitutability are particularly high, there is a unique equilibrium in which both suppliers adopt secret linear contracts. In the intermediate region, in which both supplier and retailer substitutability are intermediate, there are two equilibria, one with both suppliers adopting secret linear contracts and one with both of them adopting public two-part tariffs. Finally, in the remaining region, where supplier and retailer substitutability are low, the unique equilibrium consists of public two-part tariffs.

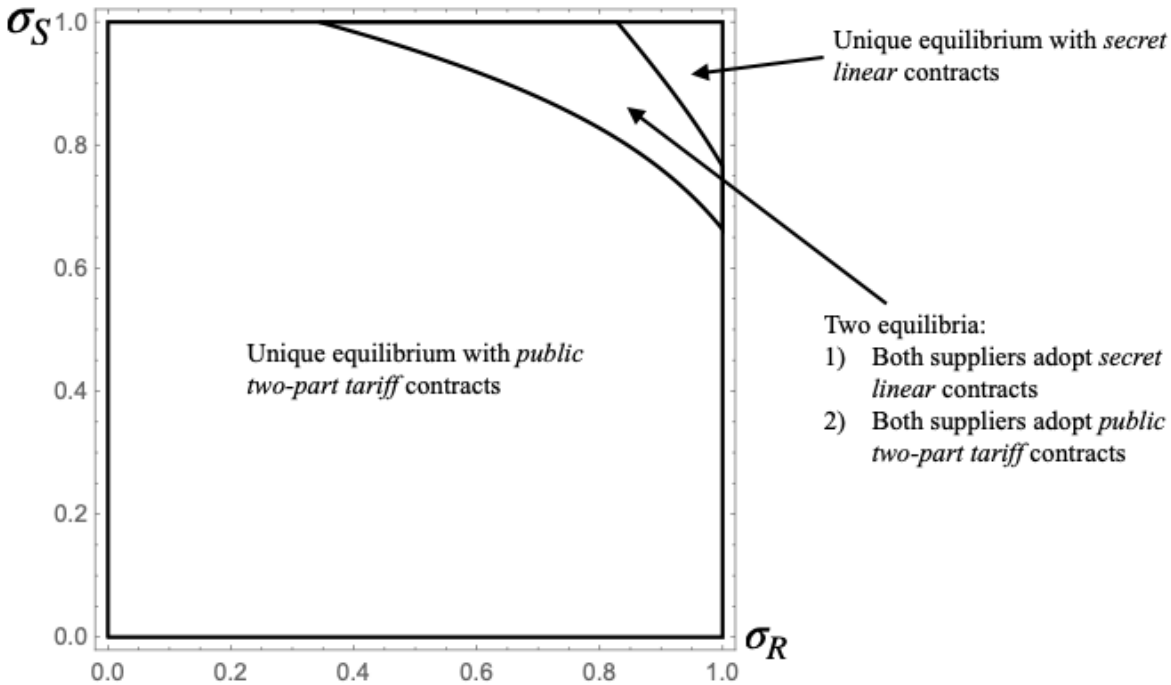

Figure 4: equilibria when all contract types available

The result that secret linear contracts can emerge as an equilibrium even when public two-part tariffs are available is striking. This result implies that for sufficiently intense competition upstream and downstream, each single supplier is willing to put itself in a competitive disadvantage by adopting secret linear contracts, involving a higher wholesale price, so as to relax upstream competition.

It is noteworthy that the condition for secret linear contracts to emerge as an equilibrium when all four contract types are available, discussed in the current section, and the condition for such contracts to be an equilibrium when only linear contracts are

available (discussed in section 3) are completely different. What the two scenarios (of fixed fees being feasible or not) have in common is that when supplier competition is intense, secret linear contracts can become particularly attractive, because then it is especially important for suppliers to relax upstream competition, and the most effective way to do so is with secret linear contracts, due to suppliers' opportunism. Yet, when all four contract types are available, *both* supplier and retailer substitutability need to be high for secret linear contracts to be adopted in equilibrium (see figure 4). This is while when only linear contracts are feasible, if supplier substitutability is high, secret contracts are adopted regardless of retailer substitutability. Even when supplier substitutability is low, secret contracts are adopted if retailer substitutability too is low (see Proposition 2 and figure 2). There are four forces that drive this difference. First, when fixed fees are feasible, rent extraction becomes a dominant consideration eliminating secret linear contracts as an equilibrium when retailer substitutability is low: each supplier wishes to defect to adopting public two-part tariffs, in order to reap retailers' high rents. Second, public two-part tariffs involve considerably lower wholesale prices than secret linear ones, especially when retailer substitutability is low, as this involves considerable double marginalization. Accordingly, the competitive advantage a supplier has when it unilaterally chooses public two-part tariffs while its rival chooses secret linear contracts is significant. Third, and relatedly, when only linear contracts are feasible, double marginalization is an issue whether or not contracts are secret. The only thing suppliers need to worry about is to what extent double marginalization is exacerbated due to secret contracts. With low retailer substitutability, secret contracts only modestly increase double marginalization. This is while when fixed fees are feasible, a supplier has the opportunity to completely eliminate double marginalization, which is exceptionally high when retailer substitutability is low. Finally, when supplier substitutability is low and retailer competition is imperfect, suppliers wish to defect to public-two part tariffs to eliminate the double marginalization caused by their margins. By contrast, when fixed fees are not available, eliminating double marginalization is not an option anyway. With low retailer substitutability, the increase in double marginalization cause by secret linear contracts is modest, so suppliers are more willing to use such contracts.

We have also verified that there are no other equilibria, neither symmetric (where both suppliers use the same type of contract) nor asymmetric (where suppliers use different contract types). For instance, both suppliers adopting public linear contracts

is never an equilibrium, because given that the other supplier adopts public linear contracts, a supplier would always defect to adopting public two-part tariffs, thereby gaining a competitive advantage, with a lower wholesale price, and being able to extract rents, via the fixed fee. Similarly, both suppliers adopting secret two-part tariffs is never an equilibrium. Secret two-part tariffs involve wholesale prices equal to marginal cost, due to opportunism (see Nocke and Rey (2018)), so although they commit a supplier to compete aggressively, they cannot constitute an equilibrium: the supplier loses the wholesale price as a tool to relax downstream competition.

In all asymmetric contract combinations but one, for all values of substitutability, at least one supplier who is not already adopting public two-part tariffs wishes to deviate to public two-part tariffs. In one contract combination, in which supplier j chooses public two-part tariffs and supplier $-j$ chooses secret linear contracts, an equilibrium fails to exist for more subtle reasons. Here, supplier $-j$, who adopted secret linear contracts, would not deviate to any other contract (including public two-part tariffs) if retailer and supplier substitutability are high. Yet, for such high retailer and supplier substitutability, supplier j (who adopted public two-part tariffs) deviates from the public two-part tariff to secret linear contracts (recall from the discussion above that secret linear contracts are the unique equilibrium if supplier and retailer substitutability are high). In other words, the combination in which supplier j adopts public two-part tariffs and supplier $-j$ chooses secret linear contracts fails to be an equilibrium because either supplier j deviates to secret linear contracts (if substitutability is high) or supplier $-j$ deviates to public two-part tariffs (if substitutability is low).

5 Managerial implications

The paper's main insight is that suppliers' choice whether to make their vertical contracts secret or public has strategic effects. When choosing whether its vertical contracts are secret or not, a supplier should consider not only how the contract directly extracts profits from the retailers or affects aggregate surplus, but also how it affects competition between suppliers. These two considerations may work in opposite directions, and the strategic effect can dominate depending on the intensity of supplier or retailer competition. In particular, our analysis yields the following managerial implications:

When contracts are linear, secret contracts can be more profitable to suppliers than public contracts Secret linear contracts result in higher wholesale prices than public linear contracts. This has the negative effect of enhancing double-marginalization, thereby reducing sales. Yet, secret linear contracts have a strategic advantage over public linear contracts in that they soften upstream competition. When, for exogenous reasons, suppliers can only adopt linear contracts, secret contracts can emerge as an equilibrium outcome when supplier substitutability is high. Even when supplier substitutability is low, if retailer substitutability is low as well, secret linear contracts again emerge in equilibrium. Yet, supplier and retailer substitutability may be such that suppliers are trapped in a prisoner's dilemma: although they would collectively earn higher profits when both of them adopt secret linear contracts, in equilibrium each of them chooses public linear contracts. These insights can help shed light on examples of linear pricing mentioned in the introduction. For instance, the linear contracts reported in Smith and Thanassoulis (2009) used in the wholesale distribution of milk to supermarket chains are predicted in our model to be secret, given that supplier substitutability is exceptionally high with milk. As for the linear contracts used at the time with printed and electronic books (as reported by Gilbert (2015)), whether contracts are expected to be secret or public in this case depends on the characteristics of the case, since substitutability between book publishers is low. Because retailer substitutability (e.g., the substitutability between Apple and Amazon, in the case of ebooks) is also quite low, our results predict these vertical contracts to be secret.. Another managerial implication of our results is that suppliers who wish to relax upstream competition via secret linear contracts need to ask their retailers to sign non-disclosure agreements. Such vertical restrictions are necessary, as we show that retailers have an incentive to make their vertical contracts transparent to each other. Hence, we should expect non-disclosure agreements to be more prevalent in industries in which, according to our results, secret linear contracts are the equilibrium.

When all contract types are available to suppliers, suppliers choose public two-part tariffs when substitutability is low and secret linear contracts when substitutability is high When all four contract types are available to suppliers, our model implies that when supplier and retailer substitutability are high, suppliers are expected to adopt secret linear contracts and when these substitutability levels are low, suppliers are expected to adopt public two-part tariffs. In future drafts, we expect to

show that these are not always the best options for suppliers. We plan to characterize regions of supplier and retailer substitutability in which suppliers are caught in a prisoner's dilemma. For instance, the equilibrium may be public two-part tariffs whereas it would be better for suppliers had they both adopted secret linear contracts. Suppliers can try to overcome such inferior equilibria by adding contractual tools. For example, Bonnet and Dubois (2010)'s structural model shows that suppliers of bottled water in France used two-part tariffs coupled with resale price maintenance. Assuming resale price-maintenance is allowed by antitrust law, it can serve as an alternative tool to relax competition. If it is prohibited, however, as is the case in many jurisdictions, including the EU, when two-part tariffs are feasible, suppliers may find themselves in an equilibrium with public two-part tariffs, while they would have preferred secret linear contracts instead. Villas Boas (2003)'s empirical study of a local yogurt market finds that her data is consistent with non-linear contracts between suppliers and retailers. Such non-linear pricing could be the result of a prisoner's dilemma among suppliers, while they would be better off with secret linear pricing, to relax upstream competition. Alternatively, suppliers and retailers in Villas Boas (2003)'s study may have been sufficiently differentiated such that public non-linear contracts become more profitable than secret linear ones.

6 Conclusion

We have studied suppliers' optimal choices of whether to keep their vertical contracts secret and linear when suppliers compete with each other and have interlocking relationships with retailers, who also compete with each other. Our analysis focused on two dimensions: first, are secret linear contracts best for suppliers' joint profits? and second, if so, do such contracts emerge in a non-cooperative equilibrium between suppliers?

We show that secret linear contracts are better for suppliers' joint profits than public linear contracts when supplier substitutability is high, or when supplier and retailer substitutability is low. Hence, competing suppliers may prefer secret to public linear contracts even though secret contracts are worse than public ones for a monopolistic supplier. The reason is strategic: although secret linear contracts enhance double marginalization and erode industry profits, they relax upstream competition. This is particularly profitable to suppliers when upstream competition is intense, i.e., when supplier substitutability is high. Even when supplier substitutability is low, secret con-

tracts can be preferred when retailer substitutability is low as well, because then the sacrifice to industry profits of relaxing upstream competition is more modest. Despite this strategic advantage of secret linear contracts we show that when supplier or retailer substitutability is low, secret linear contracts can be adopted in equilibrium only if contracts are exogenously constrained to be linear, for exogenous reasons. Absent such exogenous constraints, we show how when all four contract types are open to suppliers, with low supplier or retailer substitutability, they fail to obtain secret linear contracts in equilibrium. In future drafts, we wish to explore whether this can occur even in circumstances where both suppliers are better off with secret linear contracts. This, in turn, would imply that suppliers can benefit from exogenous factors that limit their flexibility in choosing vertical contracts. In particular, when either supplier substitutability is high, or supplier and retailer substitutability are both low, suppliers may benefit from two-part tariffs being infeasible (as is the case in many markets). Then suppliers can be able to achieve secret linear contracts in a non-cooperative game when this is their preferred contract type.

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Appendix A

Below are the proofs of the lemmas and propositions.

Proof of Lemma 1:
Differentiating the supplier's profit with respect to σ_S yields:

$$\frac{\partial \pi_S^{PL,PL}}{\partial \sigma_S} = -\frac{4(1-\sigma_S+\sigma_S^2)}{(2-\sigma_S)^3(1+\sigma_S)^2(2-\sigma_R)(1+\sigma_R)} < 0,$$

where the inequality follows because $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$. Differentiating the supplier's profit with respect to σ_R yields:

$$\frac{\partial \pi_S^{PL,PL}}{\partial \sigma_R} = \left[\frac{2(1-\sigma_S)}{(2-\sigma_S)^2(1+\sigma_S)(2-\sigma_R)^2(1+\sigma_R)^2}\right] \times (2\sigma_R - 1).$$

The term in the squared brackets is positive because $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$. The second term is positive if and only if $\sigma_R > 1/2$.

Proof of Lemma 2:
The gap in the equilibrium wholesale prices is:

$$w^{SL,SL} - w^{PL,PL} = \frac{(1-\sigma_S)\sigma_R}{(2-\sigma_S)(4-2\sigma_S+\sigma_R)}.$$

The term in the denominator is positive for all $0 \leq \sigma_S \leq 1$ and $0 \leq \sigma_R \leq 1$. The term in the nominator is positive for all $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$ and equals 0 at $\sigma_S = 1$ or $\sigma_R = 0$.

Proof of Proposition 1:
The difference between the supplier's equilibrium profit with secret linear and public linear contracts is:

$$\pi_S^{SL,SL} - \pi_S^{PL,PL} = \frac{2\sigma_R(1-\sigma_S)}{(2-\sigma_S)^2(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)(4-2\sigma_S-\sigma_R)^2}$$
$$\times \left(2\sigma_S(2-\sigma_S) - \sigma_R\right).$$

The first term is positive for all $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$. The second term is positive if and only if $\sigma_S > \overline{\sigma}_S \equiv 1 - \sqrt{1 - \frac{\sigma_R}{2}}$.

The difference between the retailer's equilibrium profit with secret linear and public linear contracts is:

$$\pi_R^{SL,SL} - \pi_R^{PL,PL} = -\frac{2\sigma_R(1-\sigma_R)(1-\sigma_S)}{(2-\sigma_S)^2(1+\sigma_S)(2-\sigma_R)^2(1+\sigma_R)(4-2\sigma_S-\sigma_R)^2}$$

$$\times\left(8-4\sigma_S-3\sigma_R+\sigma_R\sigma_S\right).$$

The first term is negative and the second term is positive for all $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$.

Proof of Lemma 4:

The gap between the equilibrium wholesale prices when supplier j chooses secret linear and supplier $-j$ chooses public linear is:

$$w_j^{SL,PL} - w_{-j}^{SL,PL} = \frac{\sigma_R(1-\sigma_S)}{2(4-\sigma_R-\sigma_S^2)},$$

which is positive for all $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$.

The gap in the equilibrium wholesale prices when both suppliers choose secret linear and when supplier j chooses secret linear while supplier $-j$ chooses public linear is:

$$w^{SL,SL} - w_j^{SL,PL} = \frac{\sigma_R\sigma_S(1-\sigma_S)}{(4-\sigma_R-\sigma_S^2)(4-2\sigma_S-\sigma_R)},$$

which is positive for all $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$.

Proof of Lemma 5:

The gap in the equilibrium profit is:

$$\pi_j^{SL,PL} - \pi^{PL,PL} = \frac{(1-\sigma_S)\sigma_R}{(2-\sigma_S)^2(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)(4-\sigma_S^2-\sigma_R)^2}$$

$$\times\left(\sigma_S^2(4-\sigma_S^2)-2\sigma_R\right).$$

The first term is positive for all $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$. The second term is positive if and only if $\sigma_S > \sqrt{2-\sqrt{4-2\sigma_R}}$.

Proof of Lemma 6:

The gap in the equilibrium profit is:

$$\pi^{SL,SL} - \pi_{-j}^{SL,PL} = \frac{(1-\sigma_S)\sigma_R}{2(1+\sigma_S)(2-\sigma_R)(1+\sigma_R)(4-2\sigma_S-\sigma_R)^2(4-\sigma_S^2-\sigma_R)^2}$$

$$\times \left(8\sigma_S^2(4-\sigma_R) - (4-\sigma_R)^2\sigma_R - 8\sigma_S^2\right).$$

The first term is positive for all $0 < \sigma_S < 1$ and $0 < \sigma_R < 1$. The second term is positive if and only if $\sigma_S > \frac{1}{2}\sqrt{2-\sqrt{4-2\sigma_R}(4-\sigma_R)}$.

Proof of Proposition 2:

The proof follows from the text and from Lemmas 5 and 6.